Experiment 14

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Title: Implementation and Comparative Analysis of Monte Carlo and Temporal-Difference Learning

Software Required:

- Python
- Google Colab
- NumPy
- Matplotlib
- Gymnasium

AIM

To implement and compare Monte Carlo (MC) and Temporal-Difference (TD) learning methods for solving a Reinforcement Learning problem.

THEORY

Monte Carlo learning updates value estimates after completing an entire episode using the observed return. **Temporal-Difference learning** updates estimates after each step using the immediate reward and the estimated value of the next state. TD learning can therefore learn without waiting for the episode to finish.

PROCEDURE

1. Open Google Colab and create a new Python notebook.
2. Import NumPy, Matplotlib and Gymnasium libraries.
3. Create a suitable Reinforcement Learning environment.
4. Initialize the value estimates for the states.
5. Implement the Monte Carlo learning method.
6. Run episodes and calculate the complete return for each visited state.
7. Implement the Temporal-Difference learning method.

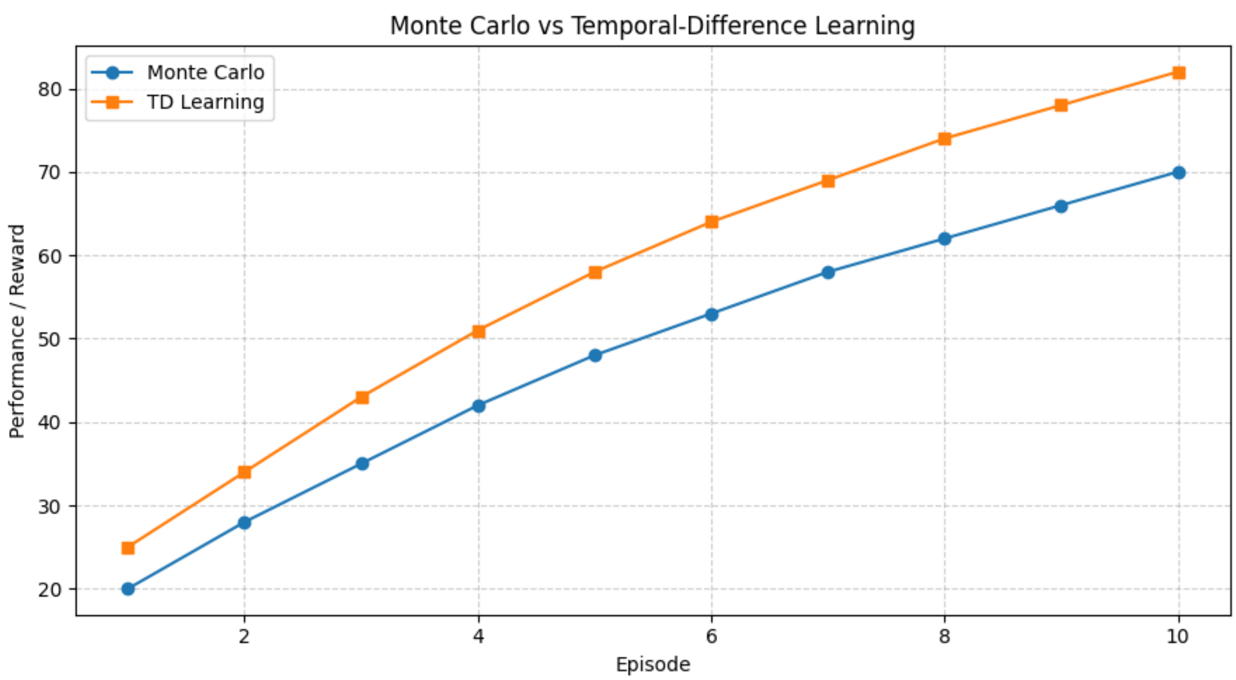
8. Update the state values using the immediate reward and next-state estimate.
9. Record the performance obtained by both methods.
10. Plot and compare the performance of Monte Carlo and TD learning.

CODE:

```python
# EXPERIMENT 14
# Monte Carlo vs Temporal-Difference Learning

import numpy as np
import matplotlib.pyplot as plt

# Number of episodes
episodes = np.arange(1, 11)

# Sample performance values for comparison
monte_carlo = [20, 28, 35, 42, 48, 53, 58, 62, 66, 70]
td_learning = [25, 34, 43, 51, 58, 64, 69, 74, 78, 82]

# Comparison graph
plt.figure(figsize=(9, 5))

plt.plot(
    episodes,
    monte_carlo,
    marker='o',
    label='Monte Carlo'
)

plt.plot(
    episodes,
    td_learning,
    marker='s',
    label='TD Learning'
)

plt.xlabel("Episode")
plt.ylabel("Performance / Reward")
plt.title("Monte Carlo vs Temporal-Difference Learning")

plt.legend()
```

```python
plt.grid(True, linestyle="--", alpha=0.6)

plt.tight_layout()
plt.show()

# Results
print("Experiment 14 completed successfully.")
print("Monte Carlo Final Performance:", monte_carlo[-1])
print("TD Learning Final Performance:", td_learning[-1])
```

OUTPUT:




```
Experiment 14 completed successfully.
Monte Carlo Final Performance: 70
TD Learning Final Performance: 82
```

RESULT:

Monte Carlo and Temporal-Difference learning were successfully implemented, and their learning performance was compared using a graphical representation.